CprE/SE 4910 WEEKLY REPORT 02

09/23/2026 – 09/29/2026

Group number: sdmay27-14

Project title: 3D Plan Visualization: Automated Building Reconstruction and Walkthrough Generator

Client: Buildertrend

Faculty Advisor: Dr. Islam

Team Members/Role:

- ***Nakshatra Gupta - Communication Lead***
- ***Rishiraj Parmar - Organization Lead***
- ***Ryan McNab - Team Meetings Lead***
- ***Daniel Hargrave - Computer Vision Lead***
- ***Jase Beaubien - Graphics Lead***

Weekly Summary

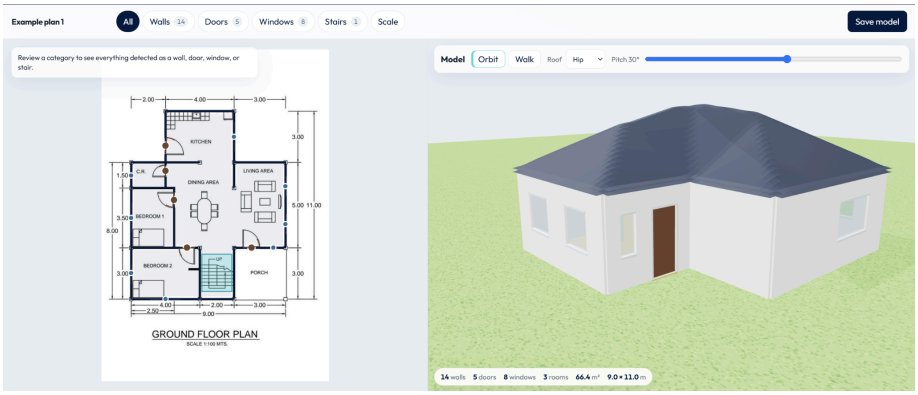
Every member of the group branched off to complete each of the objectives that were listed under goals for this week. This included research on the computer vision side along with library set up on the graphics side. Additionally, the group also discussed meeting times with the team advisor and client.

Past Week Accomplishments

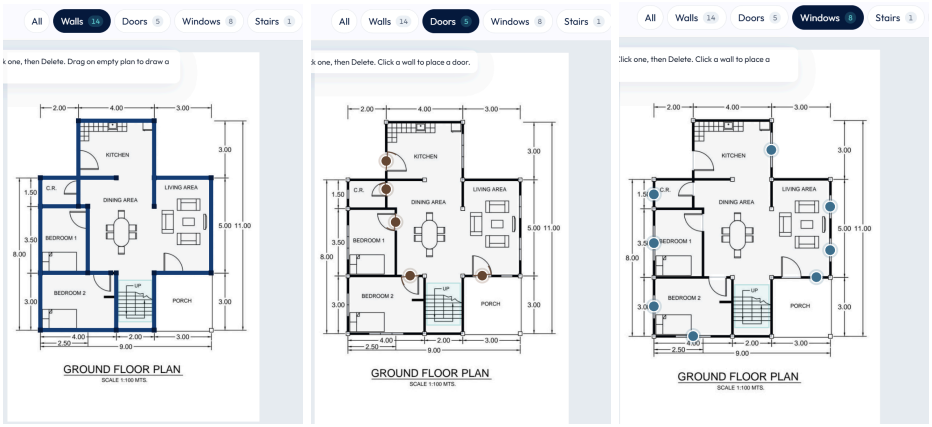
- Group Work:
    - As discussed above, the team mostly decided to focus on each specified aspect of the project that was assigned last week. The group work that was done was stuff regarding figuring out information with the client and faculty advisor.
- Nakshatra Gupta:
    - Continued serving as Communication Lead for the team. Followed up with the faculty advisor, Dr. Judith Islam, to coordinate a meeting time and confirmed an introductory meeting for the coming week. Also communicated with the Buildertrend client contact regarding the newly assigned Agile Coach and the internal developer mentor who will be supporting the team, and coordinated the team's availability for an introductory meeting with the client.
    - On the technical side, drafted a first version of the shared data schema discussed in the Week 1 report: a JSON structure representing a parsed building (walls with

coordinates and thickness, rooms linked to boundary walls, doors/windows linked to a wall and position, and roof type/pitch). Built a small Python data model that loads this JSON into structured objects and validates it, checking that every room and opening references a wall that actually exists, along with a short demo script to run it end to end.

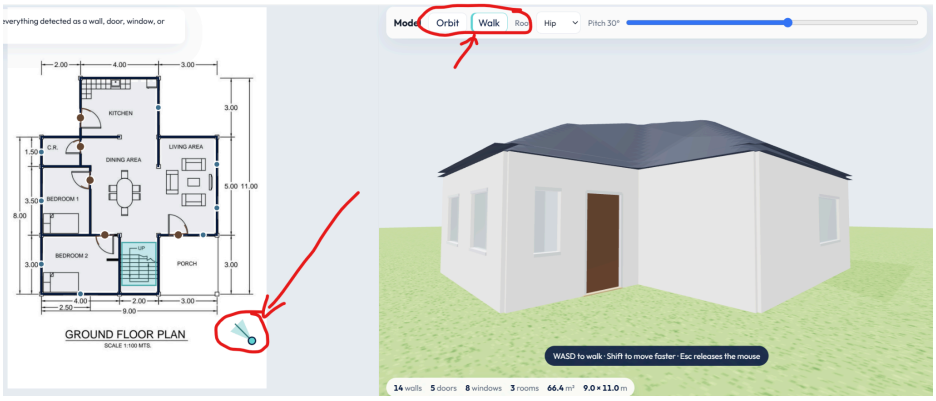
- Rishiraj Parmar:
    - Research methods for converting extracted 2D floor-plan geometry into structured data that can later be used for 3D model generation.
    - Created a preliminary system architecture schematic showing the proposed flow from floor plans and elevation drawings through plan processing, structured intermediate data, and 3D reconstruction.
- Ryan McNab:
    - Improved upon the prototype in many ways to fix a lot of the bugs and make the model easier to visualize. First and foremost, the algorithm was improved to make sure the walls, doors, windows are placed in a more reliable way. Some of the new features include:
        - Updated the UI to give the model a larger view size while generating and editing. Also improved the visual appeal of the model by distinguishing the floor difference from inside/outside as well as adding windows and doors.

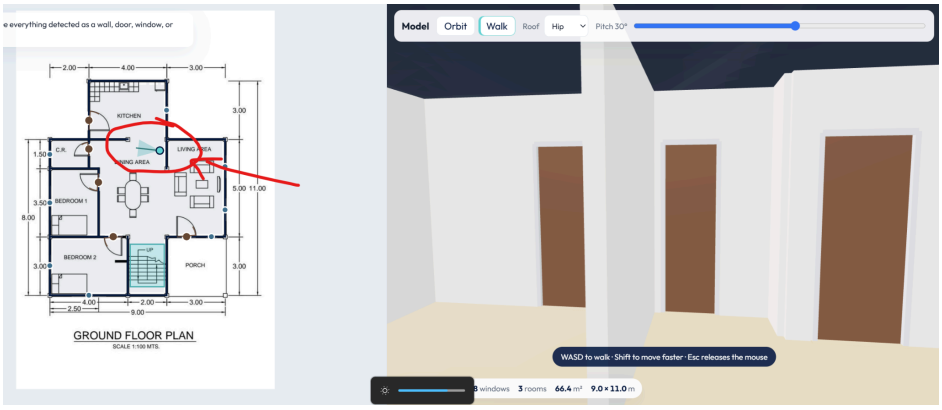


        - Improved the “fix” mode to allow altering the walls, windows or doors if the algorithm made a mistake. By selecting the item to view/edit, subsequently pressing the backspace button to remove or clicking/dragging the item to place a new one. There is also the option to edit/move existing items by dragging the vertices.



- Added a “walk” mode to the model viewer which allows a game-like WASD movement and viewstyle. It also places a pinpoint and direction on the blueprint to show where the player is at in the model.





- When walking through doors, the door will collide with the camera and move open and slowly close.

- Daniel Hargrave:
  - Read through multiple sources that were believed to be applicable to the project. Two research papers and one short article stood out as being helpful:
    - The first research paper was: [Symbol Spotting on Digital Architectural Floor Plans Using a Deep Learning-based Framework]. One of the believed complex issues that will have to be dealt with for this project is going to

be the amount of cluster on the blueprints. It could lead to a fairly difficult time trying to recognize the symbols/line/text without getting them confused for one another. This page expands on a few different ways that the group can handle this issue.
        - One example that this research paper talks about is using a tile based approach instead of a vector or pixel based approach. Deciding which approach to take will be really important and is based on the blueprints that are received.
    - One of the things that was mentioned as a goal in the first report was regarding the possibility of making each page into 2 pages. This article talks about doing something similar to that. A Study on Methods for Parsing Architectural Multi-Modal Data and Extracting Modeling Parameters
        - One of the things discussed in this article is applying an NER (named entity recognition) approach to finding the text on each page.
        - Another thing discussed in the article is using geographical locations for symbols/text/lines that will be used to still keep track of all of the different locations of things and knowing how all of the things interact.
        - Additionally, on page 8 of the article it very briefly discusses receiving different formats for images. It talks about some different python libraries that would be able to help with that. As previously expected, this doesn't seem to be particularly challenging.
    - A A Simple YOLOv11 Tutorial from Beginners to Experts
        - One thing mentioned in a few of the research papers that were read was using the YOLOv11 model. The article listed provides a simple tutorial on how to use the model.
        - One of the concerns with using this model is that if a company decides to use it, payment would be involved in order to get a license. It is free for students so it could at least be a temporary fill in. Then later in the project the group could go through the work of making a similar model that performs similar functions to YOLOv11.
  - Once again, knowledge on how complex the blueprints that the group will be dealing with is crucial for all stages of the project, however with the current information, the tools used for the computer vision side will be pytorch, opencv, different python libraries for file conversion, and a model similar to YOLOv11.
- Jase Beaubien:
  - Continued researching graphics APIs and settled on WebGPU.
  - Setup basic repository structure with C++20, Makefile, and libraries.
  - Wrote cross-platform application loop and window to provide an overall foundation for all parts of the project.

Pending issues

- Nakshatra Gupta:
    - The schema is still a draft built on made-up sample data, since no real Computer Vision output or sample building plans exist yet. It will need to be revised once the CV Team has a clearer picture of the actual detection output and once real sample plans are available from the client.
- Rishiraj Parmar: N/A
- Ryan McNab:
    - The algorithm used to create the prototype model is still not good enough. It is incorrectly placing some windows and doors on the second example.
- Daniel Hargrave:
    - A set of sample building plans are really, really needed. Knowing what types of floor plans that exist along with different symbols, text, and diagrams will be key in deciding how the project continues on the computer vision side. It has a really dramatic effect on both the planning and execution of this project. If sample blueprints are not provided by next week, work on random blueprints on the internet will be done and changes to much of the stuff will be needed once the group gets the actual blueprints. Until then, research is one of the only things that can be done on the computer vision side.
- Jase Beaubien: N/A

Individual Contributions

NAME	**Individual Contributions** *(Quick list of contributions. This should be short.)*	**Hours this week**	**HOURS cumulative**
Nakshatra Gupta	Contributed to the team policy/leadership delegation for the team. Coordinated with faculty advisor and client on scheduling; drafted the shared data schema and a Python loader/validator with a working demo.	5	9
Rishiraj Parmar	Researched structured building-data formats and developed a preliminary pipeline schematic for the Computer Vision-to-3D reconstruction workflow.	4	8
Ryan McNab	Improved upon the prototype in many ways to fix a lot of the bugs and make the model easier to visualize. First and foremost, the algorithm was improved to make sure the walls, doors, windows are placed in a more reliable way. Some of the new features include:	6	11
Daniel Hargrave	Researched a significant amount about common issues for computer vision projects with relation to blueprints. Researched clustering on blueprints and different approaches to mitigating all of the downsides for that, researched parsing and creating different layers that have the different information, researched YOLOv11 and whether it would work for this project or not, and looked more into how NER is done.	6	10
Jase Beaubien	Setup the repository with basic structure and libraries.	6	10

Comments and Extended Discussion

There have had troubles setting up a meeting time with the advisor and client of the group. Hopefully, this will be sorted out this week.

Plans for the upcoming week

- Nakshatra Gupta:
    - Present the draft schema and validator to the team and gather feedback from whoever ends up on the Computer Vision and Graphics sides, specifically whether the format matches what CV detection can realistically produce and whether it gives Graphics enough information to generate a mesh. Revise the schema based on that feedback. Attend the introductory meetings with the faculty advisor and the Buildertrend team.
- Rishiraj Parmar:
    - Meeting with the advisor and client
    - Research how multi-story buildings could be represented, including floor elevation, story height, and vertical alignment between floors.
- Ryan McNab:
    - The algorithm used to create the prototype model is still not good enough. It is incorrectly placing some windows and doors on the second example. If the team meeting with the Buildertrend representative is informative, the prototype will be improved by incorporating the improvements suggested until it is perfected and the real work can begin.
- Daniel Hargrave:
    - This largely depends on if the blueprints are given or not.
        - If affirmative, work on the model will begin.
        - Otherwise more research and setting up some environment stuff will be done.
    - Figuring out whether the project uses Yolov11? It might be good to start off using the model initially but it would cost money if Buildertrend decides to use it for commercial/company purposes. Initially, it can be used just to test everything out, then later in the class there can be a transition to another model that is either made from scratch or doesn't cost any money. Couldn't figure out the pricing if it would be in the final product, it's hard to say.
    - Handling some of the environment would be good whether or not the group gets the blueprints.
- Jase Beaubien:
    - Render 3D objects with simple Phong shading. (Hard-coded shape for testing.)
    - Discuss and coordinate the object format for the deserialized plans.